

Md. Al-Mamun Provath

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🔗 al-mamunprovath17.github.io  Google Scholar

Research Vision

My research advances **trustworthy multimodal AI** systems that seamlessly integrate visual and linguistic understanding while ensuring robustness, interpretability, and ethical alignment. I focus on developing next-generation AI that bridges theoretical innovation with real-world impact, preparing me to contribute meaningfully to leading research initiatives in responsible AI development.

Research Interests

My research advances trustworthy multimodal AI through three interconnected directions:

- **Multimodal Reasoning & LLMs:** Developing robust vision-language models with enhanced interpretability and cross-modal understanding
- **Trustworthy Computer Vision:** Ensuring fairness, security, and robustness in real-world AI deployments across healthcare and surveillance domains
- **Human-Centered AI Systems:** Bridging theoretical advances with practical applications through explainable AI and ethical design principles

Education

Chittagong University of Engineering and Technology (CUET)

Bangladesh

M.Sc. in Computer Science and Engineering

Jul 2023 – April 2025

CGPA: 3.88/4.00 (with Honors)

Thesis Focus: Violence Detection in Videos Using Deep Learning

Chittagong University of Engineering and Technology (CUET)

Bangladesh

B.Sc. in Computer Science and Engineering

Jan 2018 – Jul 2023

CGPA: 3.96/4.00 (with Honors) Merit Position: 1st among 718 students

Publications

Total Citations: 63+, i10-index: 2 (Google Scholar) — First/Co-First Author on 11 Publications — Corresponding Author() on 2 Papers*

Journal Papers

1. **Md. Al-Mamun Provath**, M. Rahman, et al. (2025). DeepGuard: Enhancing Violence Detection in Smart Cities through Deep Learning. *IEEE Access*(IF: 3.9).
2. F.H. Saikat, **Md. Al-Mamun Provath**, et al. (2025). Deep Learning-Based Image Watermarking using Catalan Transform and Non-Negative Matrix Factorization. *IEEE Access*(IF: 3.9).
3. M. Rahman, **Md. Al-Mamun Provath**, et al. (2025). CAMFusion: Context-Aware Multi-Modal Fusion Framework for Detecting Sarcasm and Humor Integrating Video and Textual Cues. *IEEE Access*(IF: 3.9).
4. **Md. Al-Mamun Provath**, K. Deb, et al. (2023). Classification of lung and colon cancer histopathological images using global context attention based convolutional neural network. *IEEE Access*(IF: 3.9).
5. M.M. Islam, **Md. Al-Mamun Provath**, et al. (2024). Strength Prediction of Geopolymer Concrete With Wide-Ranged Binders and Properties Using Artificial Neural Network. *Advances in Civil Engineering*(IF: 1.8).

Book Chapters

1. **Md. Al-Mamun Provath**, K. Deb, et al. (2023). Classification of Lung and Colon Cancer Using Deep Learning Method. In *IW-FCV*, Springer.

Selected Conference Papers

1. M.K. Islam, ..., **Md. Al-Mamun Provath*** et al. (2025). Zero-Shot Face Recognition via Fine-Tuning CLIP with Contrastive Alignment. *NCIM 2025*, pp. 27–28, Gazipur, Bangladesh.
2. **Md. Al-Mamun Provath**, K. Deb, et al. (2024). XAI-Driven Lightweight Multiscale ConvLSTM Architecture for Video Violence Detection. *IEEE ISIE*.

Research & Teaching Experience

Chittagong University of Engineering and Technology (CUET), Bangladesh

Lecturer & Research Supervisor, Department of Computer Science and Engineering Nov 2023 – Present

- **Research Mentorship:** Supervised 3 undergraduate theses, producing 1 peer-reviewed publication and 2 ongoing projects in LLM reasoning—preparing students for graduate-level research with 100% thesis completion rate
- **Teaching Excellence:** Delivered 4 courses (OOP, Compiler Design, Software Engineering, Digital Image Processing) to 400+ students

United International University (UIU), Bangladesh

Lecturer, Department of Computer Science and Engineering

Jun 2023 – Sep 2023

- Taught foundational CS courses to 250+ students achieving 95% positive feedback rating and 90% course completion rate

Enhancing Digital Government and Economy (EDGE) Project, ICT Division

Technical Mentor

Nov 2024 – Feb 2025

- Led Django Web Framework training for 50+ government developers, contributing to national digital transformation initiative

BDSET Project, Bangladesh Hi-Tech Park Authority (BHTPA), ICT Division

AI/ML Instructor

Dec 2024 – May 2025

- Delivered comprehensive training on Python, ML, and Deep Learning to 100+ participants with 98% satisfaction rating

Selective Projects [GitHub]

Academic Research Assistant – Multi-agent AI system using CrewAI and LLMs for automated literature synthesis and research proposal generation. Demonstrates advanced understanding of LLM orchestration and knowledge integration—core competencies for doctoral research in AI reasoning.

Violence Detection Framework - End-to-end deep learning pipeline achieving 94% accuracy on real-world video dataset. Showcases expertise in computer vision, and practical AI deployment in smart city applications.

Job Scraper Email Bot - Flask-based cold email generator using AI to analyze job postings and match them with relevant portfolio projects. Integrates LangChain, Groq API, ChromaDB vector database, and automated web scraping for intelligent business outreach and lead generation.

Mitochondria Detection in SEM Images - YOLOv11-based SEM image dataset and model achieving **69.4% mAP@0.5** for multi-class cellular structure detection, showcasing expertise in biomedical image analysis and deep learning.

Honors & Awards

University Gold Medal for highest CGPA in graduating class (1/718), CUET

2023

Dean's Award for Academic Excellence, CUET

2025, 2023

UGC Merit Scholarship for outstanding academic performance, Government of Bangladesh

2022

Leadership & Professional Service

Moderator, CUET Computer Club (Leading 50+ members in research initiatives)	Jul 2025–Present
Reviewer, Artificial Intelligence Review (Springer, IF: 9.6)	2025
Organizing Committee Member, IEEE ECCE Conference	Feb 2025
Research Coordinator, CUET Computer Club	2022

Technical Skills & Research Expertise

AI/ML Research (Expert): PyTorch, TensorFlow, LLM Fine-tuning, Multimodal Fusion, Transformer Architectures, GANs

Programming (Advanced): Python, Java, C/C++ **Research Tools:** Git, Linux, Weights & Biases, LaTeX, Docker

Specialized Domains: Medical Image Analysis, Video Understanding, Cross-Modal Learning, Explainable AI, Computer Vision

Data Science Pipeline: Pandas, NumPy, scikit-learn, Statistical Analysis, Large-Scale Data Processing

Selective Online Courses

Deep Learning Specialization - *Coursera*

DeepLearning.AI TensorFlow Developer Specialization - *Coursera*

Data Analyst in Python - *Dataquest.io*

References

Dr. Kaushik Deb, Professor, Dept. of CSE, CUET (debkaushik99@cu et.ac.bd)

Dr. Mohammad Shamsul Arefin, Professor, Dept. of CSE, CUET (sarefin@cu et.ac.bd)

Shayla Sharmin, Assistant Professor, Dept. of CSE, CUET (shayla_sharmin@cu et.ac.bd)